Parallel circuits

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Relate branch currents to a shared potential difference.

SUCCESS CRITERIA - I can...

1. Identify branches connected across the same two nodes.
2. Use equal voltage across parallel branches.
3. Add branch currents to find supply current.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “parallel circuit” and “node” to compare your model or result. Sentence starters are optional.

parallel circuit: branches connected across the same two nodes

node: a shared electrical connection point

current: rate of charge flow

01 Warm-up: bring back an earlier idea

A current of 3 A flows for 5 s.
Find the charge that passes.

Word help: charge = a property responsible for electric interactions

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $1/R_{total} = 1/R_1 + 1/R_2$; $I_{total} = I_1 + I_2$; $V_1 = V_2$. Parallel branches connect to the same two nodes and share voltage.

Worked reasoning: Two parallel branches of 6 and 3 ohms each have 6 V across them. Branch currents are 1 and 2 A; source current is 3 A.

Your turn: Draw the two branches and mark where the currents split and recombine.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Mark the two shared nodes in the parallel diagram. If one branch opens, can the other remain powered?
Explain.

Sentence starter: My claim is … . The diagram/data show …, so … .



Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

04 Apply the physics

A circuit board has 3 Ω and 6 Ω resistors connected in parallel across an ideal 6 V battery.
Label the voltage across each branch, calculate both currents and show how they combine at the source.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found … [unit]. This value means … in this situation.



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

05 Explain your reasoning

Why can the equivalent resistance be smaller than either branch resistance?

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A bundle of parallel rays passes through a lens and becomes 4 times as wide farther beyond it, with rays spreading immediately after the lens.
Identify the lens type and sketch the rays.
Draw the rays used in your explanation.

Word help: lens = a transparent element that changes light-ray directions

Sentence starter: My design shows … . It meets the requirement by … .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

Two parallel branches draw 0.2 A and 0.3 A.
Find the current supplied to their junction.
Label the circuit and show the current path or paths.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Return to the worked example in problem 2. Add a third 6 ohm branch to this ideal supply.
Predict each existing branch current and the new total current.
Label the circuit and show the current path or paths.

Word help: ideal = using stated simplifying conditions

Sentence starter: I predict ... . My model supports this because ... .

Draw the complete circuit and label shared nodes.

Marzano check - circle: 1 2 3